

PROBLEM STATEMENT

This memo evaluates a common decision facing retail credit card businesses:

whether to allocate incremental budget toward retaining high-value, high-churn-risk customers, or toward new customer acquisition.

Using a public credit card customer dataset, this analysis sizes the at-risk segment and tests three hypotheses about churn drivers to inform the recommendation.

APPROACH

The analysis breaks the churn problem into three MECE drivers - engagement decline, product utilization, and service friction and tests a specific, falsifiable hypothesis for each against a 10,127-customer credit card dataset.

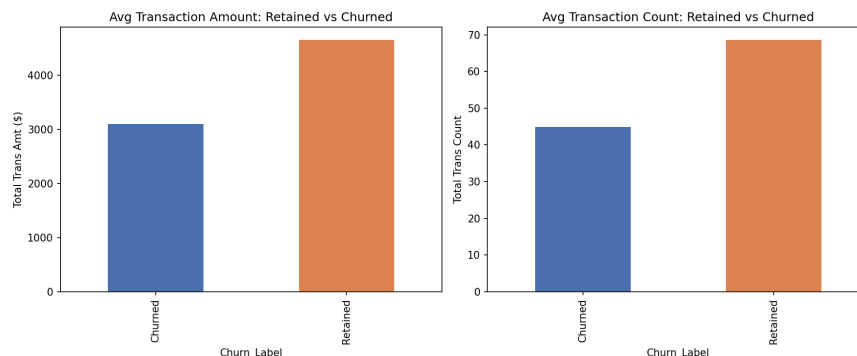
Customers are then segmented by value and churn risk to size the portion of the portfolio most exposed to attrition, informing a budget allocation recommendation between retention and acquisition spend.

MECE FRAMEWORK

1. Engagement decline - reduced transaction activity before leaving.
2. Product utilization - fewer active products held.
3. Service friction - more service contacts and inactive periods before attrition.

HYPOTHESES TESTED

H1 (Engagement decline): Churned customers show meaningfully lower transaction amount and transaction count than retained customers.

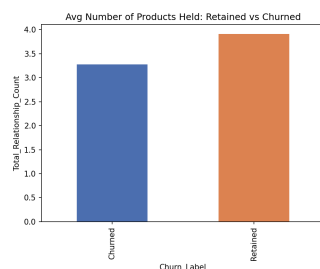


- Churned customers: ~3,095 avg transaction amount, ~45 avg transaction count

- Retained customers: ~4,655 avg transaction amount, ~69 avg transaction count

Result: Supported. Both metrics show a large, consistent gap - churned customers transact noticeably less, both in frequency and dollar volume, before leaving.

H2 (Product utilization): Churned customers hold fewer active products (Total_Relationship_Count) than retained customers.



Churned: ~3.28 products held

Retained: ~3.91 products held

Result: Partially supported. Real gap, but smaller in relative terms than H1 - a difference of about half a product, not a dramatic split.

H3 (Service friction): Churned customers show more customer service contacts and more inactive months in the 12 months prior to attrition than retained customers.

Inactive months: Churned ~2.69 vs Retained ~2.27

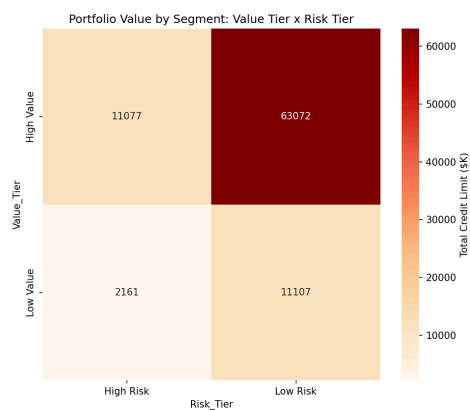
Service contacts: Churned ~2.97 vs Retained ~2.36

Result: Supported. Both move in the expected direction with a clear, visible gap in the bars.

SEGMENT SIZING

Customers were segmented into four quadrants using credit limit (value) and churn status (risk):

- **High Value / High Risk:** \$11.08M in credit exposure (12.7% of total portfolio value) - customers the bank extended the most credit to, who have already churned
- **High Value / Low Risk:** \$63.07M - the bulk of portfolio value, currently retained
- **Low Value / High Risk:** \$2.16M - smaller stakes, lower priority
- **Low Value / Low Risk:** \$11.11M - least relevant to this analysis



RECOMMENDATION

Reallocate an estimated 10-15% of new customer acquisition budget toward targeted retention offers for the High Value / High Risk segment.

- This segment represents \$11.08M in credit exposure (12.7% of total portfolio value) already lost to churn
- Even a modest reduction in attrition within this group would protect meaningful portfolio value, at a fraction of the cost of acquiring equivalent new high-value customers
- Suggested trigger points for proactive outreach: 2+ consecutive inactive months, or a service contact with no resolution follow-up

CAVEATS

- Risk tier reflects realized churn, not predictive risk - a production model would need forward-looking signals to flag at-risk customers before they leave.
- Engagement and product-utilization signals are correlated, not fully independent drivers.
- Public data used as proxy; real complaint/pricing/transaction data would sharpen this further.